

Maxim Van de Wynckel

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Education

Doctor of Sciences (PhD), Computer Science Vrije Universiteit Brussel (Elsene, Belgium) Thesis: Interoperable and Discoverable Indoor Positioning Systems	2019 - 2025
Master of Science (MSc), Applied Computer Science Vrije Universiteit Brussel (Elsene, Belgium) Graduated magna cum laude Thesis: Indoor Navigation by Centralized Tracking	July, 2019
Bridging Programme, Computer Science Vrije Universiteit Brussel (Elsene, Belgium)	2016 - 2018
Bachelor (Ba), Applied Computer Science Erasmushogeschool Brussel (Anderlecht, Belgium) Graduated magna cum laude Thesis: eID IO - Applicatie Besturing	June, 2016

Employment

Vlaanderen connect. Technical Team Lead	2025 - now
MVdW-Software Software engineering consultant	2019 - now
V.U.B. Diving Center vzw Board Member & Secretary	2025 - now
Vrije Universiteit Brussel ○ Oct, 2025 - now ○ Oct, 2019 - Sep, 2025	2019 - now External Postdoctoral Researcher Academic and Teaching Staff
ChemStream BV. Software engineer (student job)	2014 - 2018
UZ Brussel Software developer internship (student job)	2016

Publications

Journal papers (peer reviewed)

Van de Wynckel, M., & Signer, B. (2025). OpenHPS: A Modular Framework to Facilitate the Development of FAIR Positioning Systems. *Journal of Open Source Software*, 10(110), 8113. <https://doi.org/10.21105/joss.08113>

Conference papers (peer reviewed)

Van de Wynckel, M., Valadez, I., & Signer, B. (2024). FidMark: A Fiducial Marker Ontology for Semantically Describing Visual Markers. *Proceedings of The Semantic Web (ESWC 2024)*, 235–250. https://doi.org/10.1007/978-3-031-60635-9_14

Van de Wynckel, M., & Signer, B. (2023). SemBeacon: A Semantic Proximity Beacon Solution for Discovering and Detecting the Position of Physical Things. *Proceedings of the 13th International Conference on the Internet of Things, IoT '23*, 9–16. <https://doi.org/10.1145/3627050.3627060>

Van de Wynckel, M., & Signer, B. (2022). POSO: A Generic Positioning System Ontology. *The Semantic Web – ISWC 2022*, 231–247. https://doi.org/10.1007/978-3-031-19433-7_14

Van de Wynckel, M., & Signer, B. (2022). A Solid-based Architecture for Decentralised Interoperable Location Data. *Proceedings of the 12th International Conference on Indoor Positioning and Indoor Navigation (IPIN'22), CEUR Workshop Proceedings*, 3248. <https://ceur-ws.org/Vol-3248/paper11.pdf>

Van de Wynckel, M., & Signer, B. (2021). Indoor Positioning Using the OpenHPS Framework. *2021 International Conference on Indoor Positioning and Indoor Navigation (IPIN)*, 1–8. <https://doi.org/10.1109/IPIN51156.2021.9662569>

Posters (peer reviewed)

Van de Wynckel, M., & De Roock, E. (2026). OWL-SDA: A Multi-Agent System for Generating Synthetic Data to Support Ontology Evaluation. *Posters, Demos, Blue Sky, and Tutorials at SEMANTiCS 2026*. (*in press*)

Van de Wynckel, M., & Van Haute, G. (2026). ODDToolkit: An Open-Source Toolkit to Help Bridge the Developer Gap in Ontology Driven Design. *Posters, Demos, Blue Sky, and Tutorials at SEMANTiCS 2026*. (*in press*)

Van Haute, G., Goedertier, S., Fannes, P., & **Van de Wynckel, M.** (2026). Chemical Identity Lost in Regulation: A Study of Semantic Interoperability in European Chemical Substance Data. *Posters, Demos, Blue Sky, and Tutorials at SEMANTiCS 2026*. (*in press*)

Malaise, Y., **Van de Wynckel, M.**, & Signer, B. (2024). Towards Distributed Intelligent Tutoring Systems Based on User-owned Progress and Performance Data. Proceedings of the 2nd Solid Symposium (SoSy'24), 3947. <https://ceur-ws.org/Vol-3947/short3.pdf>

Van de Wynckel, M., & Signer, B. (2024). Discoverable and Interoperable Augmented Reality Environments Through Solid Pods. Proceedings of SoSy 2024 (Poster), 2nd Solid Symposium, Leuven, Belgium, May 2024, 3947. <https://ceur-ws.org/Vol-3947/short2.pdf>

Technical reports (not peer reviewed)

Van de Wynckel, M., & Signer, B. (2020). OpenHPS: An Open Source Hybrid Positioning System (WISE-2020-01). Vrije Universiteit Brussel. <https://doi.org/10.48550/ARXIV.2101.05198>

Datasets

Van de Wynckel, M., & Signer, B. (2025, May). Survey on the Privacy and Transparency of Location Data. <https://doi.org/10.5281/zenodo.15564050>

Van de Wynckel, M., & Signer, B. (2025). Sphero Dead Reckoning and CV Tracking Dataset. Kaggle. <https://doi.org/10.34740/KAGGLE/DS/6760212>

Van de Wynckel, M. (2025). Garage Positioning Dataset. Kaggle. <https://doi.org/10.34740/KAGGLE/DS/6654647>

Vermunicht, B., **Van de Wynckel, M.**, & Signer, B. (2023). 802.11 Management frames from a public location. <https://doi.org/10.5281/zenodo.8003771>

Hoebeke, N., **Van de Wynckel, M.**, & Signer, B. (2023). Object Tracking on a Monopoly Game Board. <https://doi.org/10.5281/zenodo.7990434>

Van de Wynckel, M., & Signer, B. (2021, May 9). OpenHPS: Single Floor Fingerprinting and Trajectory Dataset. <https://doi.org/10.5281/zenodo.4744380>

Press Articles

VUB researcher teaches systems that determine the location of people or objects within buildings to work together <https://press.vub.ac.be/vub-researcher-teaches-systems-that-determine-the-location-of-people-or-objects-within-buildings-to-work-together>

Jong Talent: Maxim Van de Wynckel (30) zoekt oplossingen om gebruikers meer controle te geven over hun locatiegegevens <https://www.knack.be/nieuws/technologie/jong-talent-maxim-van-de-wynckel-30-zoekt-oplossingen-om-gebruikers-meer-controle-te-geven-over-hun-locatiegegevens/>

Presentations

Public PhD Defence

2025-06-30

Venue: Vrije Universiteit Brussel, LIC0.04, Brussels, Belgium

Public PhD defence titled 'Interoperable and Discoverable Indoor Positioning Systems'

Presentation at FOSDEM 2025

2025-02-01

Venue: FOSDEM 2025, Brussels, Belgium

Presentation titled 'Discovering Indoor Environments and Positioning Systems'

Conference presentation at ESWC 2024

2024-05-28

Venue: The Semantic Web. ESWC 2024, Hersonissos, Crete, Greece

Paper presentation titled 'FidMark: A Fiducial Marker Ontology for Semantically Describing Visual Markers'

Poster presentation at SoSy 2024

2024-05-02

Venue: The 2nd Solid Symposium, Leuven, Belgium

Poster presentation titled 'Discoverable and Interoperable Augmented Reality Environments Through Solid Pods'

Conference presentation at IoT 2023

2023-11-08

Venue: The 13th International Conference on the Internet of Things, Nagoya, Japan

Paper presentation titled 'SemBeacon: A Semantic Proximity Beacon Solution for Discovering and Detecting the Position of Physical Things'

Poster presentation at BeJS conf 2023

2023-05-12

Venue: The Belgian JavaScript Conference (BeJS conf), Brussels, Belgium

Poster presentation of the OpenHPS framework

Conference presentation at ISWC 2022

2022-10-26

Venue: The 21st International Semantic Web Conference, Hangzhou, China

Paper presentation titled 'POSO: A Generic Positioning System Ontology'

Conference presentation at IPIN 2022

2022-02-06

Venue: The 12th International Conference on Indoor Positioning and Indoor Navigation, Beijing, China

Paper presentation titled 'A Solid-based Architecture for Decentralised Interoperable Location Data'

Presentation at FOSDEM 2022

2022-02-06

Venue: FOSDEM 2022, Brussels, Belgium

Presentation titled 'Rapid Prototyping of a Positioning System Using the OpenHPS Framework'

Conference presentation at IPIN 2021

2021-12-01

Venue: The 11th International Conference on Indoor Positioning and Indoor Navigation, Lloret de Mar, Spain

Paper presentation titled 'Indoor Positioning Using the OpenHPS Framework'

Teaching Experience

Courses

Next Generation User Interfaces (4018166FNR)	2019 - 2025
Level: master Titular: Prof. Dr. Beat Signer	
Web Technologies (1019888BNR)	2019 - 2025
Level: bachelor Titular: Prof. Dr. Beat Signer	
Databanken (1007156ANR)	2019 - 2025
Level: bachelor Titular: Prof. Dr. Bas Ketsman	
Web Technologies (4019862ANR)	2019 - 2023
Level: master Titular: Prof. Dr. Beat Signer	
Databases (4019814ENR)	2019 - 2022
Level: master Titular: Prof. Dr. Bas Ketsman	
Advanced Topics in Big Data (4023256FNR)	2021 - 2022
Level: master Titulars: Prof. Dr. Beat Signer, Prof. Dr. Bas Ketsman	
Advanced Topics in Information Systems (4016792ENR)	2019 - 2021
Level: master Titular: Prof. Dr. Beat Signer	
Information Visualisation (4019538FNR)	2019 - 2021
Level: master Titular: Prof. Dr. Beat Signer	
Open Information Systems (4020612FNW)	2020 - 2021
Level: master Titulars: Prof. Dr. Bart Bogaerts, Prof. Dr. Bas Ketsman	
Open Information Systems (4020612FNW)	2019 - 2020
Level: master Titular: Prof. Dr. Christophe Debruyne	

Supervised Theses

ExcerptManager: A Local-first Plugin for Excerpt-Level Scholarly Information Management Based on the Resource-Selector-Link Metamodel	2025 - 2026
Student: Zakaria Rabahi	
AI-Based Email Classification and REcommendation Using the 4D Inbox Zero Framework	2024 - 2025
Student: Seifallah Youssef	
Dynamic Data Physilication Based on Magnetic Modularity and Power Line Communication	2024 - 2025
Student: Sunidhi Singh	
Paper Explorer: A Multi-Pane Solution for Literature and Citation Exploration	2024 - 2025
Student: Mable Rajan	

A Knowledge Graph-Driven Platform for Sharing Knowledge in Data Physicalisation	2024 - 2025
Student: Lysa Roxana Kwizera	
Rethinking File Management: Designing a Human-Centric File System Interface	2024 - 2025
Student: Talha Jafar	
A Monocular SLAM Solution for the Formula Student Driverless Competition	2024 - 2025
Student: Jan Boonants	
LiDAR Indoor Map Matching for Autonomous Robots	2023 - 2024
Student: Jordan Shittu	
Identifying Devices Based on MAC Layer Properties	2022 - 2023
Student: Benjamin Vermunicht	
Heuristic Real-Time Trajectory Prediction Using Spatio-temporal Databases	2022 - 2023
Student: Ahmed Barhdadi	
Location Tracking on a Physical Game Board	2022 - 2023
Student: Nathan Hoebeke	
End-User Authoring for Location-based Implicit Human-Computer Interaction	2022 - 2023
Student: Stoffel Oostvogels	
Wi-Fi Fingerprinting: Signal Visualisation and Crowd-sourcing	2020 - 2021
Student: Ahmed Barhdadi	
Indoor Positioning by Visible Light Communication	2020 - 2021
Student: Ronald Michiels	
Anonymous Contact Tracing Fusion	2020 - 2021
Student: Yassine Bouagaz	
Comparison of SLAM Solutions with Exclusive Usage of LIDAR Observation Data	2020 - 2021
Student: Ben Bergs	
Design Guidelines for IoT Systems in Smart Companies	2020 - 2021
Student: Jeroen Grégoire	

Academic Services

Reviewer of the Semantic Web - Interoperability, Usability, Applicability (Journal)	2025
Reviewer of the Interactive, Mobile, Wearable and Ubiquitous Technologies (Journal)	2022

Reviewer of the Proceedings of the 15th International Conference on Tangible, Embedded and Embodied Interaction	2020
OAP representative of the Faculty Research Comittee Other academic staff (OAP) representative of the faculty of science and engineering.	2020 - 2025
OAP representative of the Department Council Other academic staff (OAP) representative of the department of computer science.	2019 - 2025
OAP representative of the Bachelor Program Committee Other academic staff (OAP) representative of the Bachelor program of computer science.	2019 - 2025